What Happens to the Color Vectors a, a^{b*}, c_i, $c_4^\dagger$

Explicit Treatment When Squaring and Summing

PHYS 7502 — Parker Lewis

1 What Are These Objects?

The Four Color Objects in Your Amplitude

From your diagrams, the t-channel amplitude contains four color objects:

Object	What it is	Size	Lives in
c_i	incoming quark color	3×1 column vector	fundamental rep ($\mathbf{3}$)
$c_j^\dagger$ (your $c_4^\dagger$)	outgoing quark color	1×3 row vector	conjugate rep ($\bar{\mathbf{3}}$)
a	incoming gluon color	label $a = 1, \ldots, 8$	adjoint rep ($\mathbf{8}$)
a^{b*}	outgoing gluon color	label $b = 1, \ldots, 8$	adjoint rep ($\mathbf{8}$)

1.1 Quark Color Vectors: c_i and $c_j^\dagger$

These are genuine 3-component column vectors in color space (Griffiths Eq. 9.12):

$$c_1 = r = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad c_2 = b = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad c_3 = g = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{1}$$

The outgoing quark carries the adjoint (row vector): $c_j^\dagger = c_j^T$ (since these are real). For example, $c_1^\dagger = (1 \;\; 0 \;\; 0)$.

They are orthonormal:

$$c_j^\dagger c_i = \delta_{ji} \tag{2}$$

1.2 Gluon Color States: a and a^{b*}

These are **not** 3-component vectors. Gluons live in the adjoint (octet) representation. The gluon "color state" for gluon type a is one of the 8 octet states (Griffiths Eq. 9.4):

$$\begin{aligned} |1\rangle &= (r\bar{b} + b\bar{r})/\sqrt{2} & |5\rangle &= -i(r\bar{g} - g\bar{r})/\sqrt{2} \\ |2\rangle &= -i(r\bar{b} - b\bar{r})/\sqrt{2} & |6\rangle &= (g\bar{b} + b\bar{g})/\sqrt{2} \\ |3\rangle &= (r\bar{r} - b\bar{b})/\sqrt{2} & |7\rangle &= -i(g\bar{b} - b\bar{g})/\sqrt{2} \\ |4\rangle &= (r\bar{g} + g\bar{r})/\sqrt{2} & |8\rangle &= (r\bar{r} + b\bar{b} - 2g\bar{g})/\sqrt{6} \end{aligned} \tag{3}$$

These are orthonormal: $\langle a|a'\rangle = \delta_{aa'}$. But here's the key point:

How Gluon Color Actually Enters the Amplitude

In the QCD Feynman rules, the gluon color state $|a\rangle$ doesn't directly "contract" with anything else in the amplitude. Instead, it **selects which λ^a matrix to use** at the vertex.
At a quark–gluon vertex where gluon type a is absorbed/emitted, the vertex factor is:

$$\frac{-ig_s}{2}\lambda^a_{ji}\gamma^\mu$$

The index a on λ^a IS the gluon color. The external gluon factor a_a from Griffiths' rule just tells you: "this is gluon number a, so use λ^a."
When you square and sum over gluon colors, the external state factors a_a and a^*_a combine to give $|a_a|^2 = 1$ (normalized), and the sum $\sum_{a=1}^{8}$ runs the index on λ^a.

2 Tracing Through Your t-Channel Amplitude

From your page 2, after integrating out the delta functions and setting $m = 0$:

$$M_t = \frac{-g_s^2}{4\hat{t}}\left[\bar{u}(4)\underbrace{c_j^\dagger}_{\text{row}}\underbrace{\lambda^a}_{3\times3}\gamma^\mu\varepsilon_\mu(2)\not{q}\,\gamma^\nu\varepsilon^*_\nu(3)\underbrace{\lambda^b}_{3\times3}u(1)\underbrace{c_i}_{\text{col}}\right]\underbrace{(a)(a^{b*})}_{\text{gluon states}} \tag{4}$$

Now let's separate what acts on what:

- γ^μ, $\not{q}$, γ^ν are 4×4 matrices acting on the **spinor** indices of $\bar{u}(4)$ and $u(1)$.
- λ^a, λ^b are 3×3 matrices acting on the **color** indices $c_j^\dagger$ and c_i.
- (a) and (a^{b*}) are the gluon external color states.

Since the 3×3 color matrices and 4×4 Dirac matrices act on **different spaces**, they commute. So we can factor:

$$\boxed{M_t = \frac{-g_s^2}{4}\underbrace{(c_j^\dagger\,\lambda^a\lambda^b\,c_i)}_{\text{color: a number}}\underbrace{(a)(a^{b*})}_{\text{gluon states}}\times\underbrace{\frac{1}{\hat{t}}\left[\bar{u}(4)\,\gamma^\mu\varepsilon_\mu(2)\not{q}\,\gamma^\nu\varepsilon^*_\nu(3)\,u(1)\right]}_{D_t\text{ (Dirac part)}}} \tag{5}$$

Now let's see exactly what the color piece gives us.

2.1 The Quark Color Contraction: $c_j^\dagger\,\lambda^a\lambda^b\,c_i$

This is a 1×3 row vector times a 3×3 matrix times a 3×1 column vector — the result is a **single number**:

$$c_j^\dagger\,\lambda^a\lambda^b\,c_i = \sum_{m=1}^{3}(c_j^\dagger)_m\,(\lambda^a\lambda^b)_{mi}\,c_i \quad \text{...wait, let me be more careful.} \tag{6}$$

Actually:

$$c_j^\dagger\,\lambda^a\lambda^b\,c_i = \sum_{m,n}(c_j)^*_m\,(\lambda^a)_{mn}\,(\lambda^b)_{ni}\,(c_i)_i \tag{7}$$

But c_i is the column vector with a 1 in position i and 0s elsewhere. So $(c_i)_k = \delta_{ki}$. Similarly, $(c_j)^*_m = \delta_{mj}$. Therefore:

$$c_j^\dagger \, \lambda^a \lambda^b \, c_i = \sum_{m,n} \delta_{mj} \, (\lambda^a)_{mn} \, (\lambda^b)_{ni} \, \delta = (\lambda^a)_{jn} \, (\lambda^b)_{ni} = \sum_n (\lambda^a)_{jn} (\lambda^b)_{ni} \tag{8}$$

Quark Color Vectors Collapse to Matrix Elements

$$\boxed{c_j^\dagger \, \lambda^a \lambda^b \, c_i = (\lambda^a \lambda^b)_{ji}} \tag{9}$$

The quark color column vectors c_i and $c_j^\dagger$ simply **pick out the** (j, i) **element** of the matrix product $\lambda^a \lambda^b$. After this step, the quark color vectors are gone — replaced by a number.

Example: If the incoming quark is red ($i = 1$) and the outgoing quark is blue ($j = 2$), then:

$$c_2^\dagger \, \lambda^a \lambda^b \, c_1 = (0 \;\; 1 \;\; 0) \, \lambda^a \lambda^b \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = (\lambda^a \lambda^b)_{21} \tag{10}$$

2.2 The Gluon Color States: $(a)(a^{b*})$

The incoming gluon is type a with state $|a\rangle$. The outgoing gluon is type b with state $|b\rangle^*$.

In the amplitude M_t, these are just **labels** telling you which λ matrix to use. The factor $(a)(a^{b*})$ doesn't contract with anything else in the amplitude — it's already encoded in the indices on λ^a and λ^b.

So for a **specific** gluon color configuration (specific values of a and b):

$$M_t\big|_{\text{specific } a,b} = \frac{-g_s^2}{4} \, (\lambda^a \lambda^b)_{ji} \;\times\; D_t \tag{11}$$

The gluon color states (a) and (a^{b*}) have done their job: they told us to use λ^a at the p_2 vertex and λ^b at the p_3 vertex.

3 Squaring: What Happens to Each Color Object

For a specific color configuration (i, j, a, b):

$$|M_t|^2 = \frac{g_s^4}{16} \, |(\lambda^a \lambda^b)_{ji}|^2 \;\times\; |D_t|^2 \tag{12}$$

Now let's see what happens when we sum/average.

3.1 Step 1: $|M_t|^2$ Explicitly

Write out $|(\lambda^a \lambda^b)_{ji}|^2$:

$$|(\lambda^a \lambda^b)_{ji}|^2 = (\lambda^a \lambda^b)_{ji} \, \left[(\lambda^a \lambda^b)_{ji}\right]^* \tag{13}$$

The complex conjugate of a matrix element: $[(\lambda^a \lambda^b)_{ji}]^* = [(\lambda^a \lambda^b)^\dagger]_{ij} = [(\lambda^b)^\dagger (\lambda^a)^\dagger]_{ij}$.

Since the λ matrices are Hermitian ($\lambda^{a\dagger} = \lambda^a$):

$$[(\lambda^a \lambda^b)_{ji}]^* = (\lambda^b \lambda^a)_{ij} \tag{14}$$

So:

$$|(\lambda^a \lambda^b)_{ji}|^2 = (\lambda^a \lambda^b)_{ji} \, (\lambda^b \lambda^a)_{ij} \tag{15}$$

3.2 Step 2: Sum Over Outgoing Quark Color j

The outgoing quark can be any of 3 colors. Using $c_j^\dagger c_j = 1$ for each j:

$$\sum_{j=1}^{3} (\lambda^a \lambda^b)_{ji} \, (\lambda^b \lambda^a)_{ij} = \left[(\lambda^a \lambda^b)(\lambda^b \lambda^a) \right]_{ii} \tag{16}$$

This is the (i, i) element of the matrix product $(\lambda^a \lambda^b)(\lambda^b \lambda^a)$.

3.3 Step 3: Sum Over Incoming Quark Color i (and Average)

$$\frac{1}{3} \sum_{i=1}^{3} \left[(\lambda^a \lambda^b)(\lambda^b \lambda^a) \right]_{ii} = \frac{1}{3} \operatorname{Tr}\left[\lambda^a \lambda^b \lambda^b \lambda^a \right] \tag{17}$$

Quark Colors → Trace

$$\boxed{\frac{1}{3} \sum_{i,j=1}^{3} \left| c_j^\dagger \, \lambda^a \lambda^b \, c_i \right|^2 = \frac{1}{3} \operatorname{Tr}\left[\lambda^a \lambda^b \lambda^b \lambda^a \right]} \tag{18}$$

The quark color vectors c_i and $c_j^\dagger$ are completely gone. Summing over all quark colors turned their contribution into a **trace** over 3×3 color matrices.

3.4 Step 4: Sum Over Gluon Colors a and b (and Average Over a)

Now we sum over the outgoing gluon color $b = 1, \ldots, 8$ and average over the incoming gluon color $a = 1, \ldots, 8$:

$$\mathcal{C} = \frac{1}{8} \sum_{a=1}^{8} \sum_{b=1}^{8} \frac{1}{3} \operatorname{Tr}\left[\lambda^a \lambda^b \lambda^b \lambda^a \right] \tag{19}$$

Gluon Colors → Sums Over λ Indices

$$\boxed{\mathcal{C} = \frac{1}{24} \sum_{a=1}^{8} \sum_{b=1}^{8} \operatorname{Tr}\left[\lambda^a \lambda^b \lambda^b \lambda^a \right]} \tag{20}$$

The gluon color states (a) and (a^{b*}) are completely gone. Summing over all gluon colors just means running the indices a and b on the λ matrices from 1 to 8.

3.5 Step 5: Evaluate the Trace

Inner sum first: $\displaystyle\sum_{b=1}^{8} \lambda^b \lambda^b = \frac{16}{3} \mathbb{I}_3$ (quadratic Casimir).

Substitute:

$$\mathcal{C} = \frac{1}{24} \sum_a \operatorname{Tr}\left[\lambda^a \cdot \frac{16}{3} \mathbb{I} \cdot \lambda^a \right] = \frac{16}{72} \sum_a \operatorname{Tr}(\lambda^a \lambda^a) = \frac{16}{72} \times 16 = \frac{256}{72} = \frac{32}{9} \tag{21}$$

4 The Complete Chain: From Color Vectors to a Number

Here is the full sequence of what happened to every color object:

Complete Chain

In the amplitude M_t:

$$M_t \ \ni \ c_j^\dagger \, \lambda^a \, \lambda^b \, c_i \ \ (a)(a^{b*})$$

Step 1 — Quark vectors pick out matrix elements:

$$c_j^\dagger \, \lambda^a \lambda^b \, c_i = (\lambda^a \lambda^b)_{ji} \qquad \text{(a number for each } i, j, a, b)$$

Step 2 — Square:

$$|(\lambda^a \lambda^b)_{ji}|^2 = (\lambda^a \lambda^b)_{ji} (\lambda^b \lambda^a)_{ij} \qquad \text{(used Hermiticity)}$$

Step 3 — Sum j, then sum+avg i:

$$\frac{1}{3} \sum_{i,j} (\lambda^a \lambda^b)_{ji} (\lambda^b \lambda^a)_{ij} = \frac{1}{3} \mathrm{Tr}[\lambda^a \lambda^b \lambda^b \lambda^a] \qquad \text{(quark colors} \to \text{trace)}$$

Step 4 — Sum b, then sum+avg a:

$$\frac{1}{24} \sum_{a,b} \mathrm{Tr}[\lambda^a \lambda^b \lambda^b \lambda^a] \qquad \text{(gluon labels} \to \text{sums on } \lambda \text{ indices)}$$

Step 5 — Casimir + trace identity:

$$\sum_b \lambda^b \lambda^b = \frac{16}{3} \mathbb{I}, \quad \mathrm{Tr}(\lambda^a \lambda^a) = 2 \quad \Longrightarrow \quad \frac{1}{24} \cdot \frac{16}{3} \cdot 16 = \frac{32}{9}$$

Bottom line: All four color objects (c_i, $c_j^\dagger$, a, a^{b*}) have been consumed. What's left is a single number: $\boxed{\mathcal{C} = 32/9}$.

5 Final Assembly

$$\overline{|\mathcal{M}|^2}_{\mathrm{QCD}} = \underbrace{\frac{g_s^4}{16}}_{\text{coupling}} \times \underbrace{\frac{32}{9}}_{\text{all colors eaten}} \times \underbrace{\left[-2\left(\frac{\hat{s}}{\hat{t}} + \frac{\hat{t}}{\hat{s}}\right)\right]}_{\text{your QED Dirac result}} \tag{22}$$

$$= \frac{2g_s^4}{9} \times \left[-2\left(\frac{\hat{s}}{\hat{t}} + \frac{\hat{t}}{\hat{s}}\right)\right] = \frac{-4g_s^4(\hat{s}^2 + \hat{t}^2)}{9\,\hat{s}\hat{t}} \tag{23}$$

Final Cross Section

$$\boxed{\frac{d\hat{\sigma}}{d\hat{t}} = \frac{-4\pi\alpha_s^2(\hat{s}^2 + \hat{t}^2)}{9\,\hat{s}^3\hat{t}}} \tag{24}$$